

SC-900

Learning path: *2*
Describe the capabilities
of Microsoft Entra



Learning path agenda



- Describe the function and identity types of Microsoft Entra ID.
- Describe the authentication capabilities of Microsoft Entra ID.
- Describe the access management capabilities of Microsoft Entra.
- Describe the identity protection and governance capabilities of Microsoft Entra.

Module 1: Describe the function and identity types of Microsoft Entra ID



Module 1 introduction

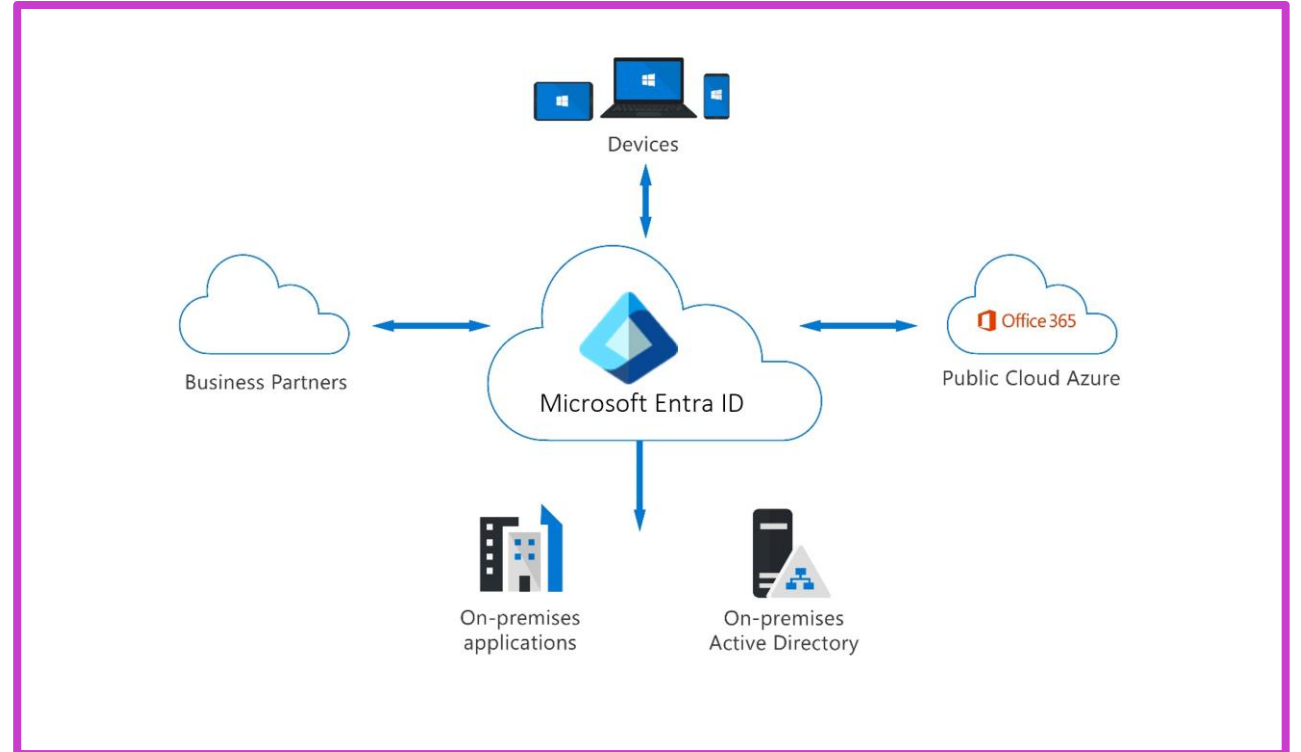
After completing this module, you'll be able to:

- 1** Describe the core functionality of Microsoft Entra ID.
- 2** Describe the types of identities supported by Microsoft Entra ID.
- 3** Describe the concept of hybrid identity as supported by Microsoft Entra ID.

Microsoft Entra ID

Microsoft's cloud-based identity and access management service.

- Organizations can enable their employees, guests, and others to sign in and access the resources they need.
- Provide a single identity system for their cloud and on-premises applications.
- Protect user identities and credentials to meet an organization's access governance requirements.
- Subscribers to Azure services, Microsoft 365, or Dynamics 365 automatically have access to Microsoft Entra ID.
- Identity secure score.



Identity types

Human (user) identities

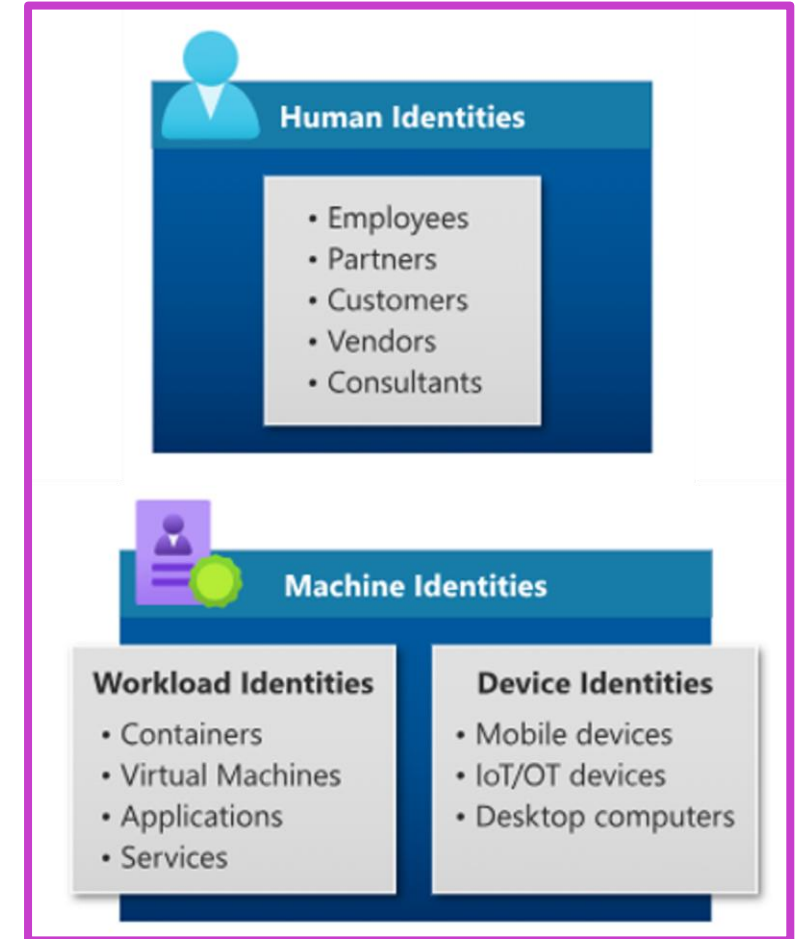
- Internal users – Employees.
- External users – Guests, partners, customers, and so on.

Workload identities (an identity assigned to an application or service)

- Service principal – Uses Microsoft Entra ID for identity and access functions; app developers manage credentials.
- Managed identities – A service principal managed in Microsoft Entra ID that eliminates the need for app developers to manage credentials.

Devices

- Microsoft Entra ID registered – Support for bring your own device.
- Microsoft Entra ID joined – Device joined via an organizational account.
- Hybrid joined – Devices are joined to your on-premises Active Directory and Microsoft Entra ID, requiring organizational account to sign in.



Demo

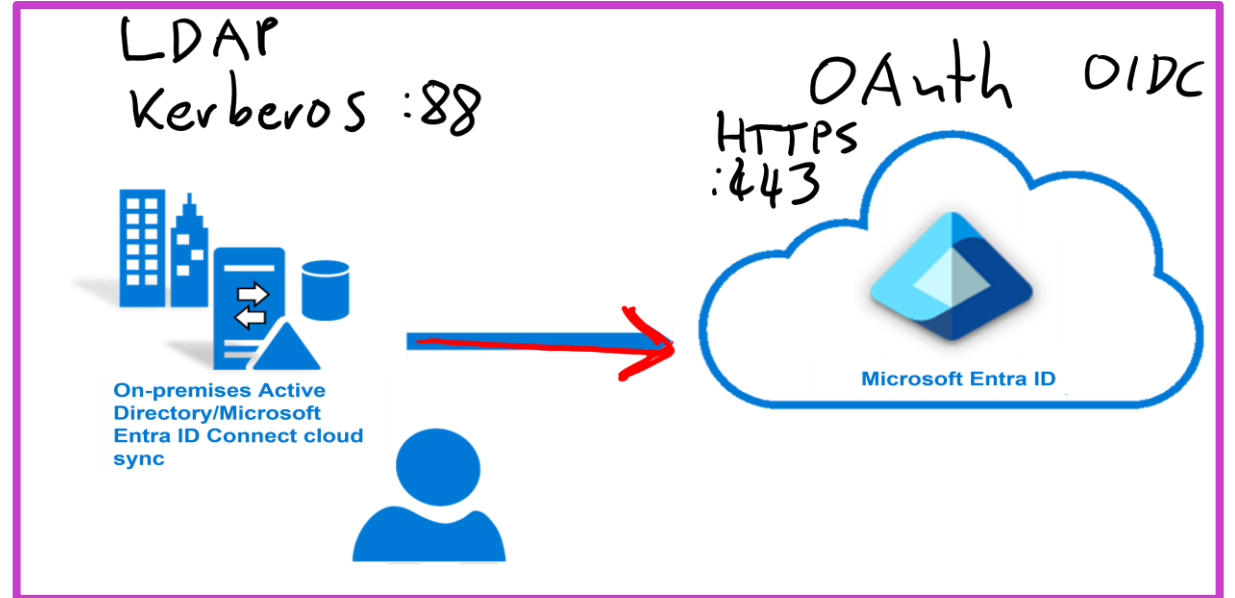
Microsoft Entra ID user settings



Hybrid identity

What is a hybrid identity?

- A common user identity for authentication and authorization to on-premises and cloud resources.
- Hybrid identity is accomplished through:
 - Inter-directory provisioning – A user in Active Directory is provisioned into Microsoft Entra ID.
 - Synchronization – Identity information for your on-premises users and groups matches the cloud.
- Microsoft Entra ID Connect cloud sync – A method for provisioning and synchronization.



Module 2: Explore the authentication capabilities of Microsoft Entra



Module 2 introduction

After completing this module, you'll be able to:

- 1** Describe the authentication methods of Microsoft Entra ID.
- 2** Describe multifactor authentication in Microsoft Entra ID.
- 3** Describe the password protection and management capabilities of Microsoft Entra ID.

Authentication methods of Microsoft Entra

Passwords (primary auth)

Phone-based authentication

- SMS (primary and secondary auth)
- Voice (secondary auth)

OATH (secondary auth)

- Standard for how one-time password codes are generated
- SW tokens
- HW tokens

Passwordless (primary and secondary auth)

- Windows Hello
- Microsoft Authenticator
- FIDO2
- Certificates (primary auth)

Bad: Password	Good: Password and...	Better: Password and...	Best: Passwordless
123456 qwerty password iloveyou Password1	 SMS  Voice	 Authenticator (Push Notifications)  Software Tokens OTP  Hardware Tokens OTP (Preview)	 Authenticator (Phone Sign-in)  Window Hello  FIDO2 security key  Certificates

FIDO2

Multifactor authentication (MFA)

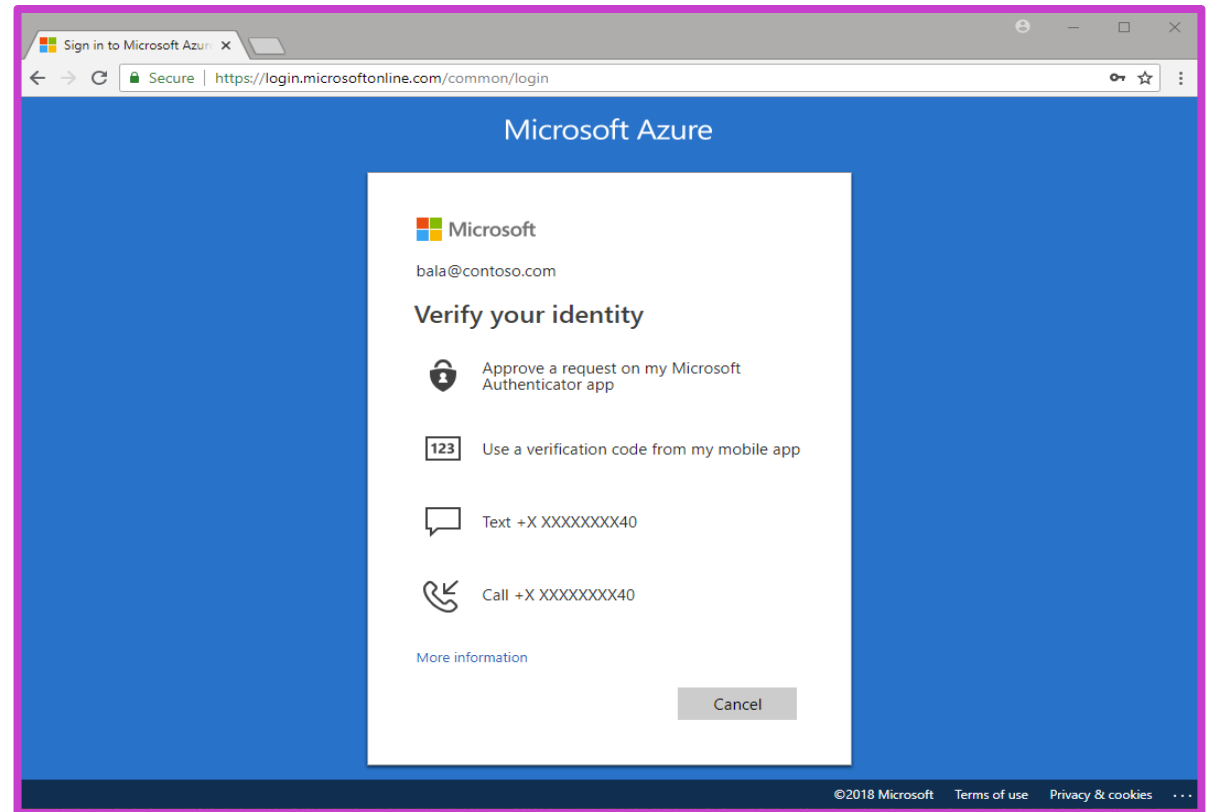
Dramatically improves the security of an identity, while still being simple for users.

MFA requires more than one form of verification

- Something you know.
- Something you have.
- Something you are.

Security defaults

- Requires all users to complete MFA as needed.
- Forces administrators to use MFA.
- Enforces MFA for all users.



Demo

Authentication methods and MFA



Password protection and management capabilities

Reduce the risk of users setting weak passwords:

- Global banned password list.
- Custom banned password lists.
- Protecting against password spray.
- Integrates with an on-premises Active Directory environment.

The screenshot shows the 'Password protection for Windows Server Active Directory' settings. It includes fields for 'Custom smart lockout' (Lockout threshold: 10, Lockout duration in seconds: 60), a toggle for 'Enforce custom list' (set to Yes), and a text area for 'Custom banned password list' containing the following text: contoso, fabrikam, tailwind, michigan, wolverine, harbaugh, howard. At the bottom, there is a toggle for 'Enable password protection on Windows Server Active Directory' (set to Yes).

Custom smart lockout	
Lockout threshold ⓘ	10
Lockout duration in seconds ⓘ	60
Custom banned passwords	
Enforce custom list ⓘ	☒ Yes ☐ No
Custom banned password list ⓘ	contoso fabrikam tailwind michigan wolverine harbaugh howard
Password protection for Windows Server Active Directory	
Enable password protection on Windows Server Active Directory ⓘ	☒ Yes ☐ No

Module 3: Explore the access management capabilities of Microsoft Entra



Module 3 introduction

After completing this module, you'll be able to:

- 1** Describe Conditional Access and its benefits.
- 2** Describe Global Secure Access.
- 3** Describe Microsoft Entra ID roles and role-based access control (RBAC).

Conditional Access

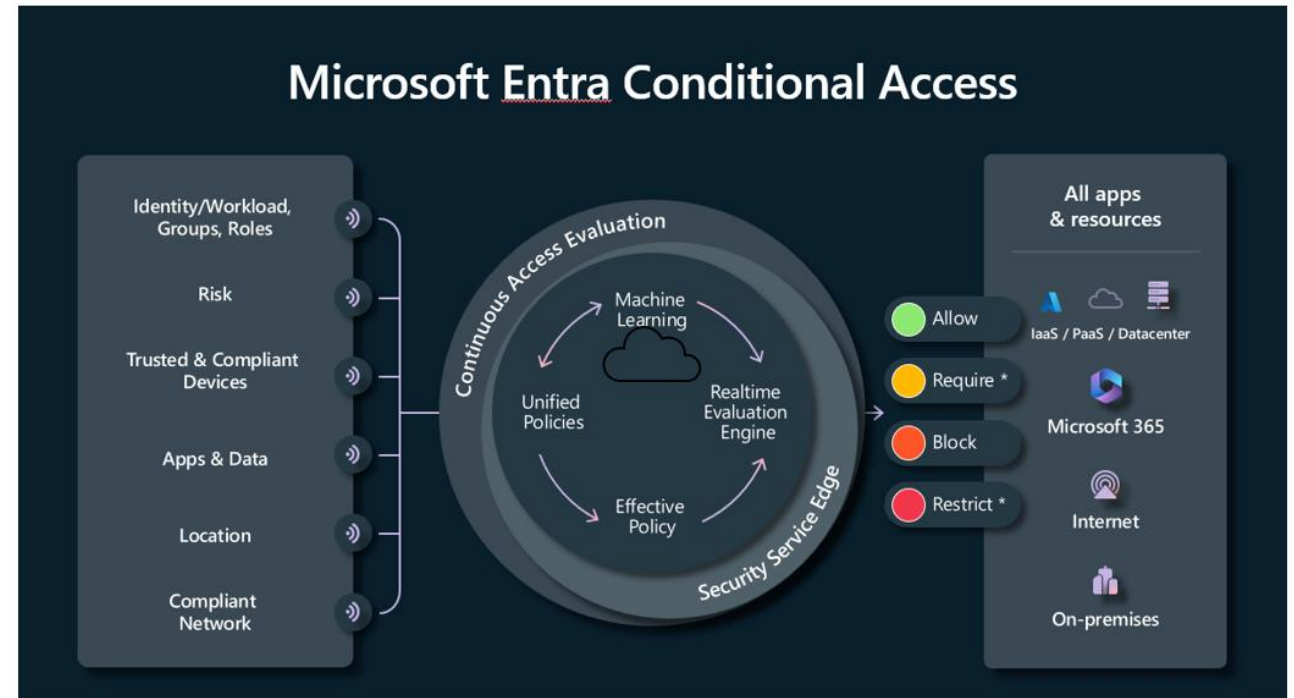
At their simplest, Conditional Access (CA) policies are if-then statements.

Assignments determine which signals to use

- Users, groups, workload identities, directory roles.
- Cloud apps or actions.
- Sign-in and user risk detection.
- Device or device platform.
- IP location.
- More...

Access controls determine how a policy is enforced

- Block access.
- Grant access – Require one or more conditions to be met before granting access.
- Session control – Enable a limited experience.



Demo

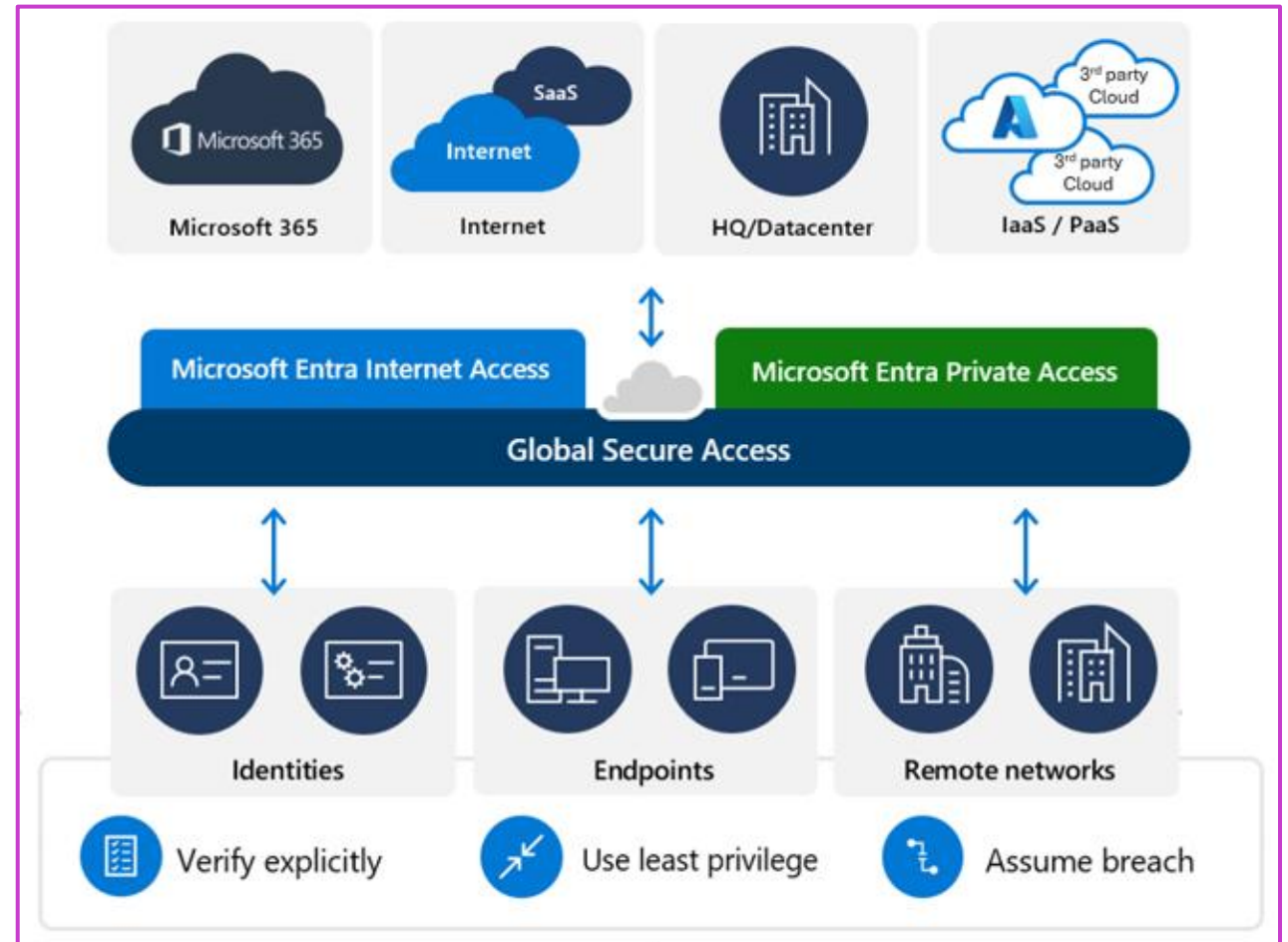
Microsoft Entra Conditional Access



Microsoft Entra Global Secure Access

GSA converges **Zero Trust network, identity, and endpoint access controls** to secure access to any app or resource, from any location, device, or identity.

- **Microsoft Entra Internet Access** secures access to SaaS applications, including Microsoft Services, and public internet apps.
- **Microsoft Entra Private Access** provides your users secure access to your private, corporate resources.

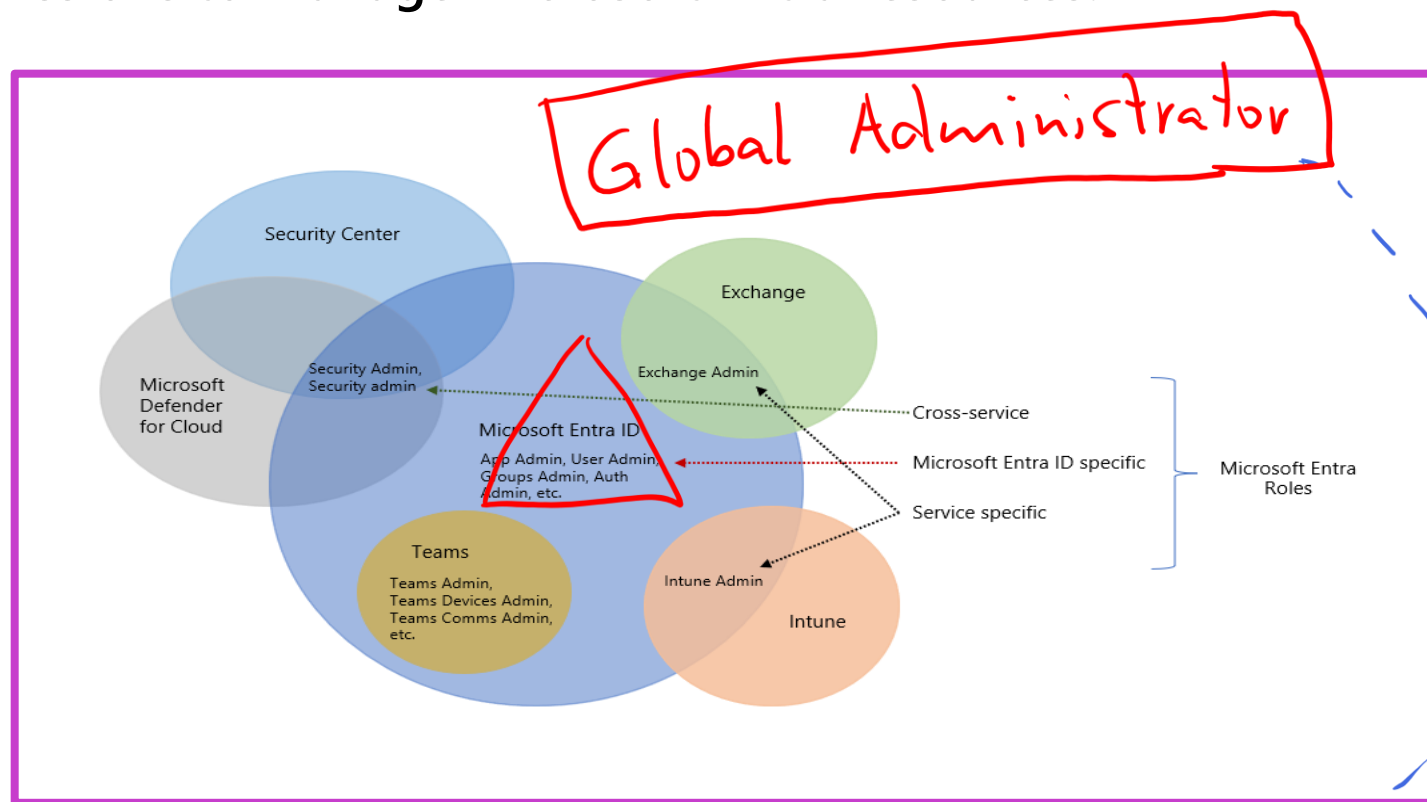


2x

Microsoft Entra roles and role-based access control (RBAC)

Microsoft Entra ID roles control permissions to manage Microsoft Entra resources.

- Built-in roles.
- Custom roles.
- Categories of Microsoft Entra roles:
 - Microsoft Entra specific
 - Service-specific
 - Cross service
- Only grant the access users need.



Azure

owner

Module 4: Describe the identity protection and governance capabilities of Microsoft Entra



Module 4 introduction

After completing this module, you'll be able to:

- 1** Describe the identity governance capabilities of Microsoft Entra.
- 2** Describe Privileged Identity Management (PIM).
- 3** Describe the capabilities of Microsoft Entra Identity Protection.
- 4** Describe Microsoft Entra integration with Microsoft Security Copilot.

Identity governance in Microsoft Entra

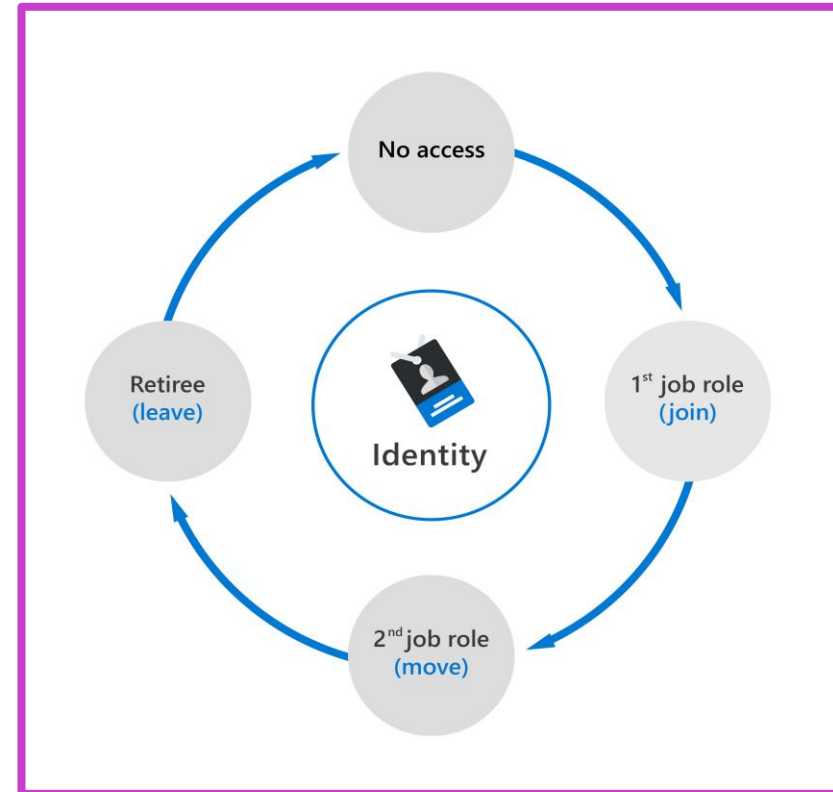
The right people have the right access to the right resources.

The tasks of Microsoft Entra identity governance

- Govern the identity life cycle.
- Govern access life cycle.
- Secure privileged access for administration.

Identity life cycle

- Join: A new digital identity is created.
- Move: Update access authorizations.
- Leave: Access may need to be removed.



Access reviews

Access reviews

- Enable organizations to efficiently manage group memberships, access to enterprise applications, and role assignment.
- Ensure that only the right people have access to resources.
- Used to review and manage access for both users and guests.

Multistage access reviews

- Support up to three review stages.
- Support workflows to meet recertification and audit requirements calling for multiple reviewers.
- Reduce the number of decisions each reviewer is accountable for.

Contoso

Please review users' access to the Finance Web app in FrickelsoftNET

Sarah Hoelzel, your organization requested that you approve or deny continued access for one or more users to the **Finance Web** app in the **FinanceWeb** access review. The review period will end on **September 5, 2020**.

Hi FinanceWeb team - please review the list of users who can access your FinanceWeb application. Help us remove any unwanted access from users that no longer work with the app. More information:

<https://finweb.contoso.com/access/reviews>

Start review >

Learn how to [perform an access review](#) and more about [Azure Active Directory access reviews](#).

[Privacy Statement](#)

Microsoft Corporation, One Microsoft Way, Redmond, WA 98052

Facilitated by



Privileged Identity Management (PIM)

PIM enables you to manage, control, and monitor access to important resources in your organization.

- 1 Just in time, providing privileged access only when needed, and not before.
- 2 Time-bound, by assigning start and end dates that indicate when a user can access resources.
- 3 Approval-based, requiring specific approval to activate privileges.
- 4 Visible, sending notifications when privileged roles are activated.
- 5 Auditable, allowing a full access history to be downloaded.

Microsoft Entra Identity Protection

Detect

- User risk
- Sign-in risk

Investigate

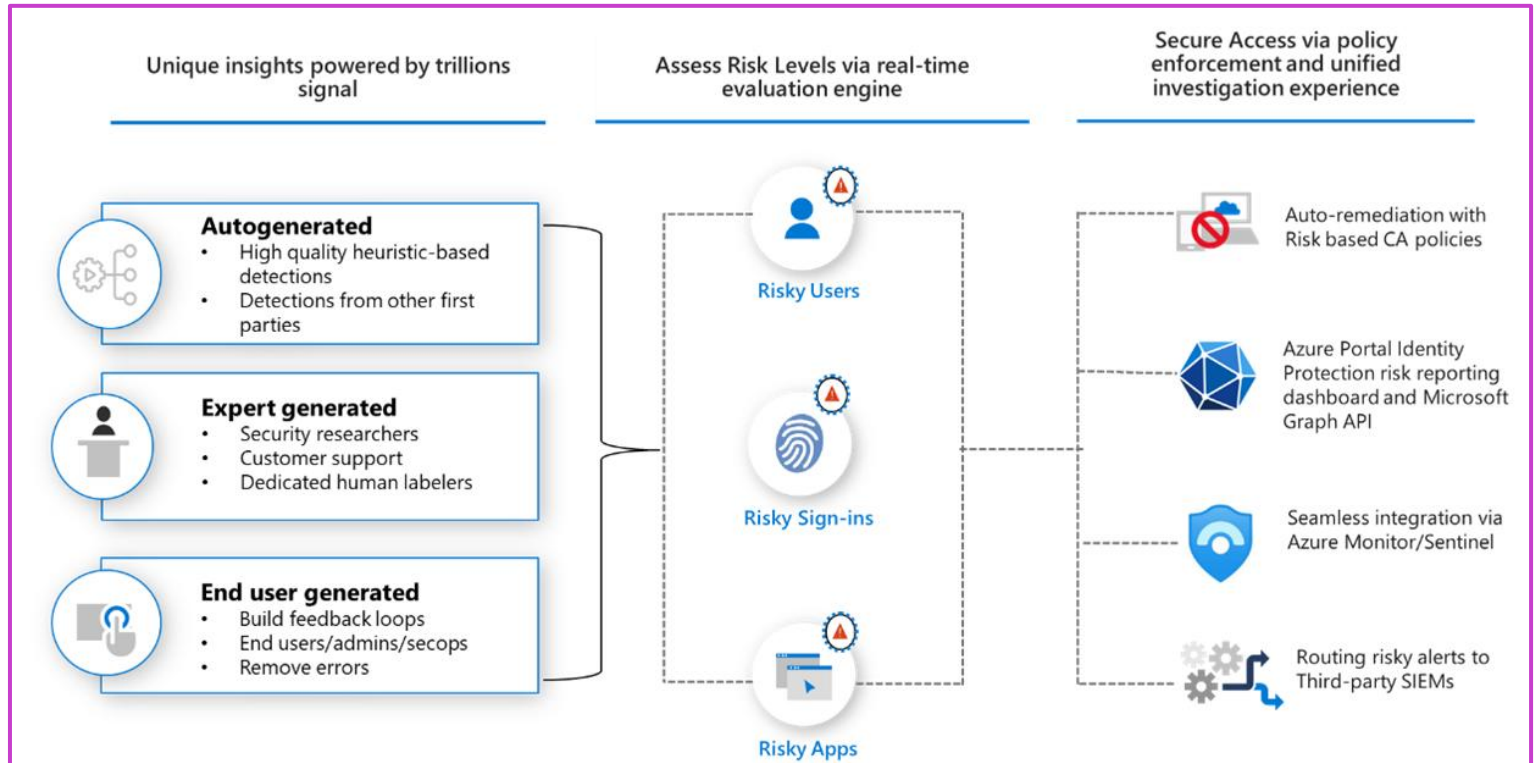
- Risk detections report
- Risky sign-ins report
- Risky users report (embeds Copilot)
- Risky workload identities report

Remediate

- Automated remediation
- Manual remediation

Export

- Export risk detection data to first and third-party utilities for further analysis.



Microsoft Entra integration with Microsoft Security Copilot



Standalone experience:

- Capabilities in standalone experience are built-in prompts.
- Use natural language to create your own prompts.

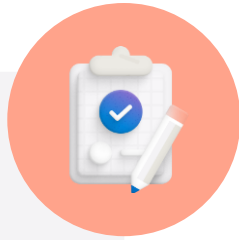
Embedded experience:

- Supported in Risky users' report.
- Summarize a user's risk level, provide insights, and provide recommendations for rapid mitigation.

A screenshot of the Microsoft Security Copilot interface showing a list of actions related to Microsoft Entra. The title 'MICROSOFT ENTRA' is highlighted with a red box. The list includes:

- Explore a summary of a users active risk with Entra ID Protection.**
View a detailed summary of a Microsoft Entra ID users risk.
- Explore diagnostic log collection in Microsoft Entra**
View settings for diagnostic log collection and streaming of activity logs in Microsoft Entra ID
- Explore Microsoft Entra audit log details**
View changes to applications, groups, users, and licenses in Microsoft Entra ID
- Find group details in Microsoft Entra**
View Microsoft Entra ID group ownership and membership details
- Find sign-in logs in Microsoft Entra**
View Microsoft Entra ID sign-in log details including policy evaluation results, and details on the loc...
- Find user details in Microsoft Entra**
View contact information, authentication method registration, and account details for users in Micr...
- Investigate identity risks with Entra ID Protection**
View details of Microsoft Entra ID users with high, medium, or low risk of compromise

Learning path summary



Describe the capabilities of Microsoft Entra.

In this learning path, you have:

- Learned about Microsoft Entra ID and the services and identity types it supports.
- Explored the authentication capabilities of Microsoft Entra and MFA.
- Explored the access management capabilities of Microsoft Entra, with Conditional Access and Microsoft Entra RBAC.
- Described identity protection and governance capabilities of Microsoft Entra, including PIM and access reviews.
- Learned about the capabilities of Microsoft Entra Identity Protection.

Knowledge check



An organization has completed a full migration to the cloud and has purchased devices for all its employees. All employees sign in to the device through an organizational account configured in Microsoft Entra ID. Select the option that best describes how these devices are set up in Microsoft Entra ID.

- A. These devices are set up as Microsoft Entra ID registered.
- B. These devices are set up as Microsoft Entra ID joined.
- C. These devices are set up as Hybrid Microsoft Entra ID joined.

After hearing of a breach at a competitor, the security team wants to improve identity security within their organization. What should they implement to provide the greatest protection to user identities?

- A. Multifactor authentication.
- B. Require security questions for all sign-ins.
- C. Require strong passwords for all identities.

Knowledge check continued



An organization plans to implement Conditional Access. What do admins need to do?

- A. Create policies that enforce organizational rules.
- B. Check that all users have multi-factor authentication enabled.
- C. Amend your apps to allow Conditional Access.

An organization has recently conducted a security audit and found that four people who have left were still active and assigned global admin roles. The users have now been deleted but the IT organization has been asked to recommend a solution to prevent a similar security lapse happening in future. Which solution should they recommend?

- A. Entitlement management.
- B. Identity Protection.
- C. Privileged Identity Management.

